AIRFOIL DESIGN CHALLENGE

F1 Rear Wing Aerodynamic Problem-Round 1 Submission

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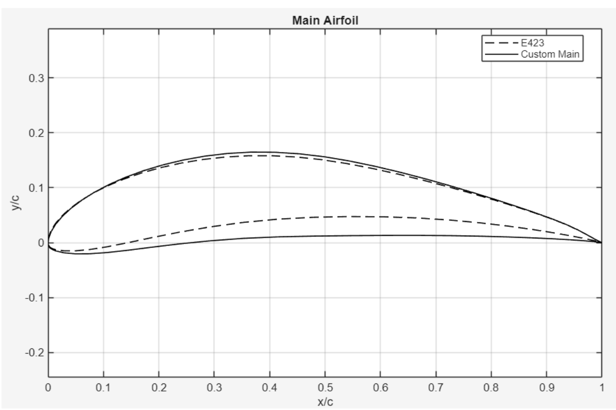
[CAD LINK 1] [CAD LINK 2]

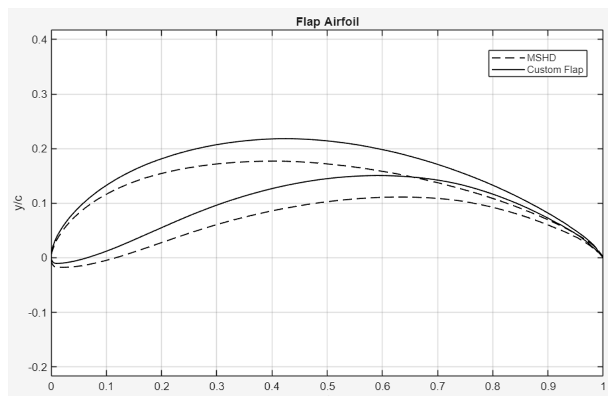
1. Airfoil Profile Selection & Custom Generation

To fully capitalize on the high-downforce requirements of circuits like Monaco and Hungary, specially custom made airfoils were designed. The main and flap airfoils used for the rear wing are inspired from the airfoils Eppler-423 and MSDH (Motor Sport High Downforce) respectively. They were then modified heavily to suit the maximum downforce and limited drag requirement of the problem statement.

2. Custom Main Airfoil (Derived & XFOIL Optimized)

The standard E423 is a high-lift, low-Re airfoil, which generally suffers from leading-edge stall at high Angles of Attack (AoA). For this F1 application, I adjusted the camber location, shifting the point of maximum camber significantly aft (towards the trailing 50% of the chord). This "aft-loading" forces the pressure recovery zone backward, ensuring the boundary layer remains attached to the suction surface even at the extreme 15.5° AoA setting. Furthermore, a custom sine-wave thickness distribution was iteratively applied to bring the maximum thickness to 16.05% of the 220mm chord. This satisfies the 15–20% rule while providing the necessary structural stiffness at the wing core without choking the spanwise airflow.





3. Custom Flap (Derived & XFOIL Optimized)

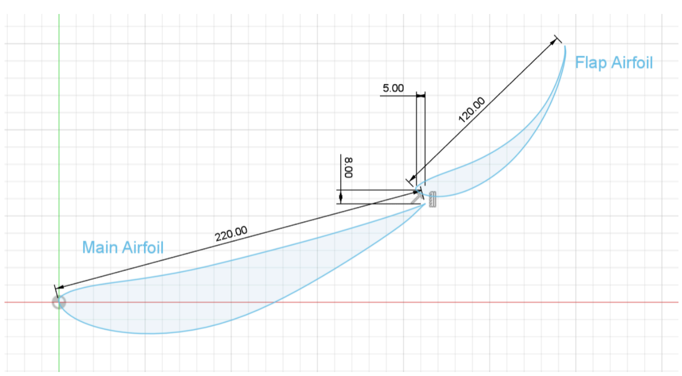
The flap is meant to be a high-energy recovery tool. While inspired by the motor sport high downforce airfoil, the baseline camber is a little too passive for a 27° relative deployment. An aggressive natural camber scaling was mathematically applied to the aft 30% of the flap, simulating the aerodynamic effect of a Kármán-Trefftz trailing edge (a virtual Gurney flap). This extreme curvature enforces a massive Kutta condition angle, ejecting flow violently upwards. To prevent stall and drag-inducing separation, the trailing edge spline was mathematically smoothed to prevent self-intersection. Thickness was strictly held at 12.87%, complying with the 10–14% rule.

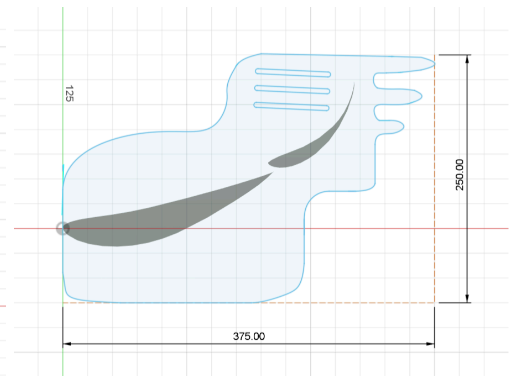
4. Dimensional Compliance & Angle of Attack (AoA) Strategy

* Main Element Chord (220 mm) & Flap Chord (120 mm): Both chords were maximized to the absolute limit. A larger surface area dictates a larger physical low-pressure zone, delivering the maximum raw downward vertical force allowable by the regulations.

* Total Span (1050 mm): Maximized to increase the wing's Aspect Ratio. A higher aspect ratio fundamentally reduces induced drag, ensuring straight-line overtaking speeds are maintained despite the colossal downforce.

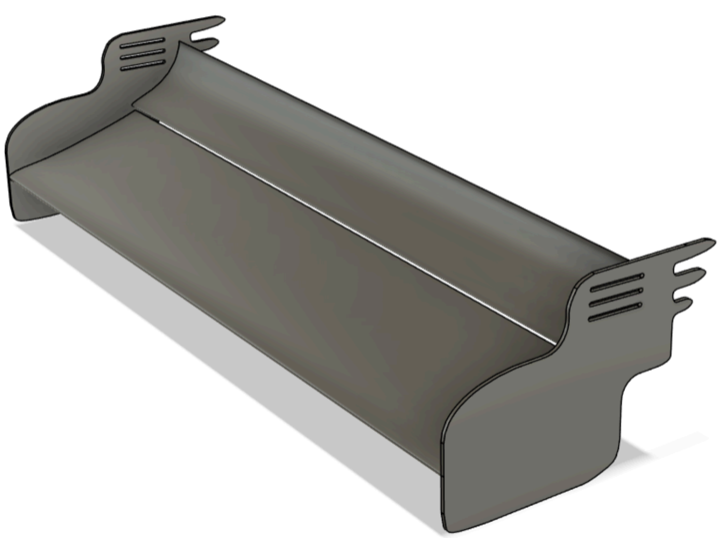
* Main Element AoA (15.5°): Pushing to the 18° limit would risk catastrophic flow separation prior to the slot gap in yaw or turbulent dirty air. A 15.5° angle guarantees robust flow attachment to the custom aft-loaded suction surface while perfectly feeding the flap.
* Flap AoA (27° Relative): Maximizing the flap to ~27° relative to the main plane acts as a violent air-ejector. Because of the aggressive camber applied, this generates an immense suction peak directly beneath the car's rear axis.
* Slot Gap Kinematics: To maintain separation control over the 27° flap, the flap's leading edge was aggressively positioned with a 4 mm horizontal overlap and an 8 mm vertical slot gap. This configuration mathematically functions as a high-pressure nozzle, injecting a high-velocity jet from the pressure side tangentially into the flap's boundary layer to permanently lock the flow to its surface.

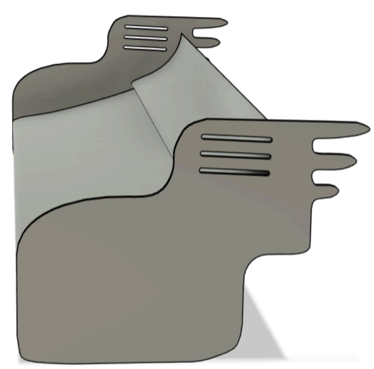




5. End Plate Geometry & Slot Rationale

The endplate design deviates heavily from symmetric traditional fencing. Since the problem statement clearly stated that it is not compulsory for the CAD to follow all dimensional constraints, I have ignored the end plate width constraint for this particular design to allow more structural integrity of the flap airfoil. Three slanted "stretched-cylinder" slots were introduced into the upper-rear quadrant of the endplate, immediately adjacent to the flap's high-pressure leading zone. These slots allow the violent high-pressure air trapped by the gap to bleed out horizontally through the fencing in a highly controlled manner. Instead of shedding one massive drag-inducing wingtip vortex, the slots dissolve the vortex into several smaller threads, rescuing straight-line maximum speed while abandoning zero downforce in the corners.





CAD LINKS

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